Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

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TE - 1	**Batch: T11**	**Sem: 5**	**Roll No:12**	**Date:20/8/2026**

EXPERIMENT 05

Aim To set up a Continuous Integration (CI) pipeline using Jenkins and GitHub to build, test, and deploy a simple HTML application automatically.

LO Mapped : 03

LO : To set up a Continuous Integration Pipeline using Jenkins/GitHub Act.

Theory

1. Continuous Integration & Continuous Delivery (CI/CD)

- **CI Core Concept:** Developers push updates into a shared repository frequently. Automated systems pull the latest commit to compile, test, and validate changes instantly, reducing integration friction and eliminating "integration hell" at the end of a release cycle.
- **CD Pipeline:** Extends CI by ensuring that validated code is automatically deployed to testing, staging, or production environments, shortening the software feedback loop.

2. Core Components of Jenkins Architecture

- **Jenkins Controller (Master):** The central node responsible for managing user configurations, scheduling build jobs, dispatching tasks to agents, and displaying pipeline metrics.
- **Jenkins Agents (Nodes):** Worker instances (local host or external servers) that execute the tasks specified in build scripts.
- **Plugins:** Modules that extend Jenkins functionality to integrate with version control systems (Git, GitHub), build tools, containers, and notification engines.

3. Jenkins Pipeline as Code & Declarative Syntax

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- **Pipeline as Code:** Storing build configuration files (Jenkinsfile) directly inside the project repository ensures version tracking, peer code review, and auditability for deployment scripts alongside application code.
- **Declarative Pipeline:** A domain-specific language (DSL) based on Groovy syntax that structures workflows clearly using strict directives:
    - **pipeline**: The root block containing the complete execution definition.
    - **agent**: Specifies where the pipeline executes (e.g., any, dedicated node label, or Docker container).
    - **stages**: Encloses individual pipeline phases.
    - **stage**: Groups steps logically into targeted actions (e.g., Build, Test, Deploy).
    - **steps**: Contains the executable terminal commands or plugins (e.g., echo, bat, sh).

4. Source Code Management (SCM) & Authentication

- **Git & Remote Repositories:** Version control tracks atomic commits locally and synchronizes them with remote providers like GitHub.
- **Personal Access Token (PAT):** A secure, scoped alternative to account passwords required by GitHub for HTTPS terminal authentication and automated server access.
- **SCM Polling & Webhooks:** Mechanisms used by Jenkins to detect SCM changes. Polling periodically queries the remote repository, whereas webhooks notify Jenkins instantly upon every git push.

Implementation

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```
C:\Users\HP 15L>java --version
openjdk 25.0.4 2026-07-21 LTS
OpenJDK Runtime Environment Temurin-25.0.4+7 (build 25.0.4+7-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4+7 (build 25.0.4+7-LTS, mixed mode, sharing)

C:\Users\HP 15L>mkdir jenkins-html-demo

C:\Users\HP 15L>cd jenkins-html-demo

C:\Users\HP 15L\jenkins-html-demo>echo "<html><body><h1>Hello Jenkins CI/CD!</h1></body></html>" > index.html

C:\Users\HP 15L\jenkins-html-demo>echo pipeline { agent any; stages { stage('Build') { steps { echo 'Building the application...' } } stage('Test') { steps { echo 'Testing th
e application...' } } stage('Deploy') { steps { echo 'Deploying the application...' } } } } > Jenkinsfile

C:\Users\HP 15L\jenkins-html-demo>git init
Initialized empty Git repository in C:/Users/HP 15L/jenkins-html-demo/.git/

C:\Users\HP 15L\jenkins-html-demo>git add .

C:\Users\HP 15L\jenkins-html-demo>git commit -m "Initial commit with app and Jenkinsfile"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'HP 15L@805-35.(none)')

C:\Users\HP 15L\jenkins-html-demo>git config --global user.email "bhatiarashi19@gmail.com"

C:\Users\HP 15L\jenkins-html-demo>git branch -M main

C:\Users\HP 15L\jenkins-html-demo>git remote add origin https://github.com/bhatiarashi19-ops/jenkins-html-demo.git
```

```
C:\Users\HP 15L\jenkins-html-demo>git push -u origin main
remote: Repository not found.
fatal: repository 'https://github.com/bhatiarashi19-ops/jenkins-html-demo.git/' not found

C:\Users\HP 15L\jenkins-html-demo>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 24 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 436 bytes | 436.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/bhatiarashi19-ops/jenkins-html-demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\HP 15L\jenkins-html-demo>type C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword
111c567fdef94b97a810c6f23c351c74

C:\Users\HP 15L\jenkins-html-demo>git add Jenkinsfile

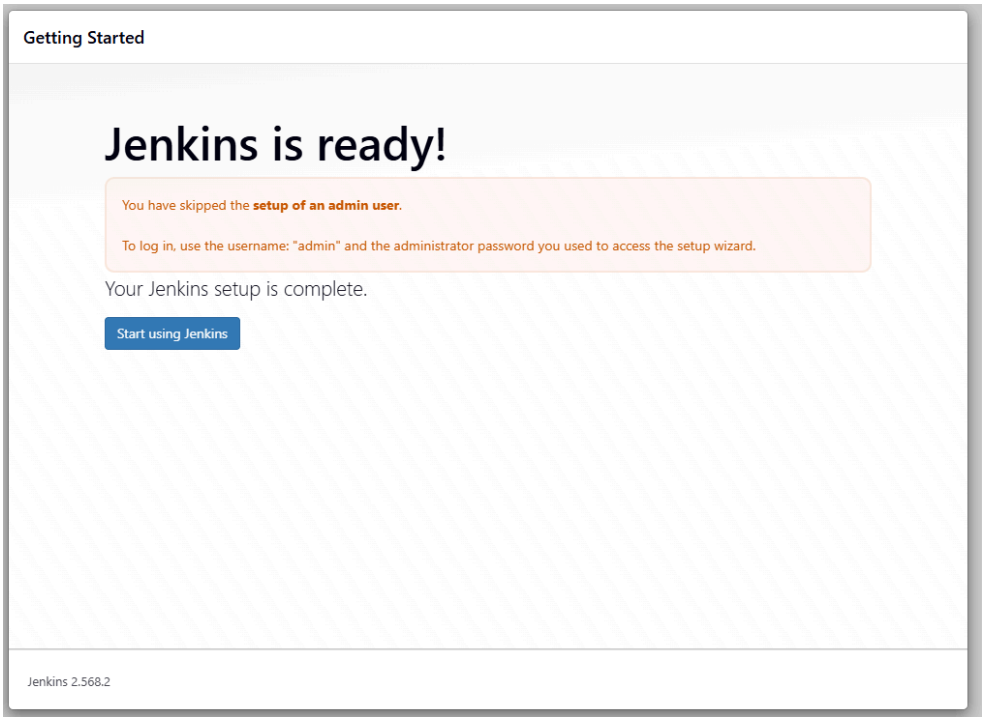
C:\Users\HP 15L\jenkins-html-demo>git commit -m "Fix Jenkinsfile formatting"
[main b235af6] Fix Jenkinsfile formatting
 1 file changed, 20 insertions(+), 1 deletion(-)

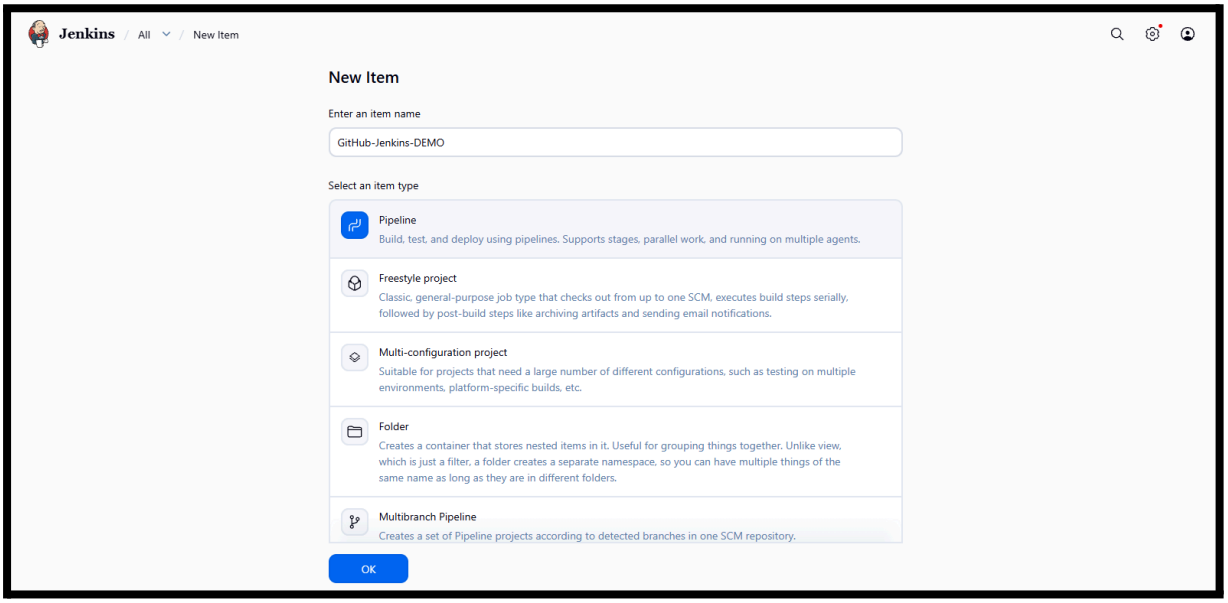
C:\Users\HP 15L\jenkins-html-demo>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 24 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 408 bytes | 408.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/bhatiarashi19-ops/jenkins-html-demo.git
   e280654..b235af6  main -> main

C:\Users\HP 15L\jenkins-html-demo>
```

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Output :





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Conclusion Experiment No. 5 was completed successfully. A simple HTML project was initialized locally, connected to GitHub via Personal Access Token authentication, and configured within Jenkins using a declarative Jenkinsfile. The pipeline automatically fetched source code, parsed multi-stage instructions, executed the Build, Test, and Deploy tasks sequentially, and achieved a successful build status (Finished: SUCCESS).